Do not write outside the box

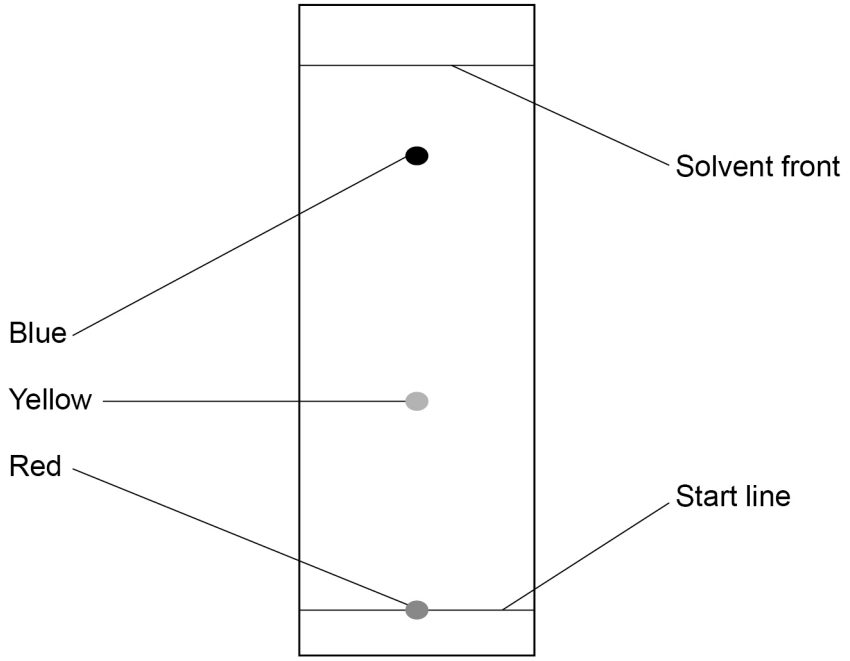
0 4 Printer ink is a mixture of chemicals.

A student used chromatography to investigate the colours in a printer ink.

The student put a spot of the printer ink on the start line.

Figure 4 shows the results.

Figure 4



Not to scale

0 4 . 1 Explain why the red colour did **not** move from the start line.

[2 marks]

_____

_____

_____

_____

Question 4 continues on the next page

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0 4 . 2 The blue colour moved 6.4 cm up the chromatogram.

The R_f value of the blue colour is 0.87

Calculate the distance moved by the solvent.

Give your answer to 2 significant figures.

[4 marks]

Distance moved by the solvent (2 significant figures) = _____ cm

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There were four colours in the printer ink.

0 4 . 3 Suggest **one** reason why only three colours were visible on the chromatogram.

[1 mark]

0 4 . 4 Suggest how the student could use chromatography to show there were four colours in the printer ink.

[1 mark]

8

Turn over for the next question